Lecture 26: Cylinders and Quadric Surfaces

Lecturer: Chng Zhi Yee (zy_chng@um.edu.my)

Based on Thomas' Calculus, 15th Edition in SI Units, Pearson Education Limited

Sem II, 2025/2026

Learning Outcome

At the end of this lecture, you should be able to:

- ☐ Identify cylinders based on their equation.
- ☐ Identify standard quadric surfaces based on their equations.

Cylinders

A cylinder is a surface generated by moving a straight line along a given planar curve while holding the line parallel to a given fixed line.

Can be represented by an equation missing one of the variables.

Example:
Parabolic cylinder $y = x^2$ Circular cylinder $x^2 + y^2 = 1$

Level Curves

A level curve (or contour) of a function is a curve parallel to xy-plane corresponding to the equation of the form $f(x, y) = c$, where c is a constant.

For each level curve, the corresponding value of c gives the height of the curve above (or below) the xy-plane.

Example: Sketch the level curves for the function $f(x, y) = x^2 + y^2$.

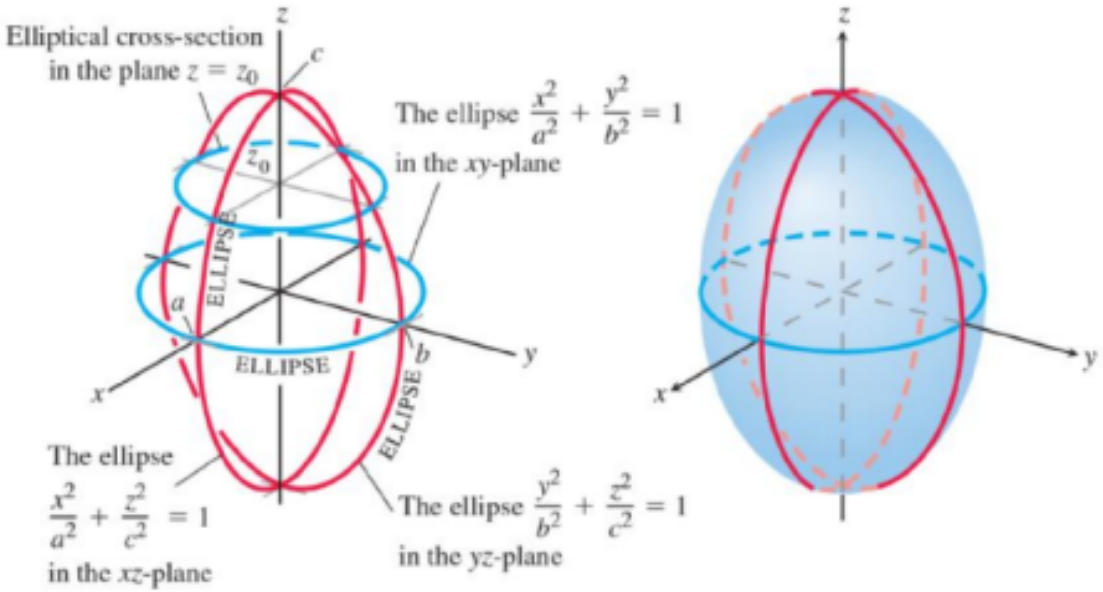
Quadric Surfaces

A quadric surface is the graph in space of a equation of the general form

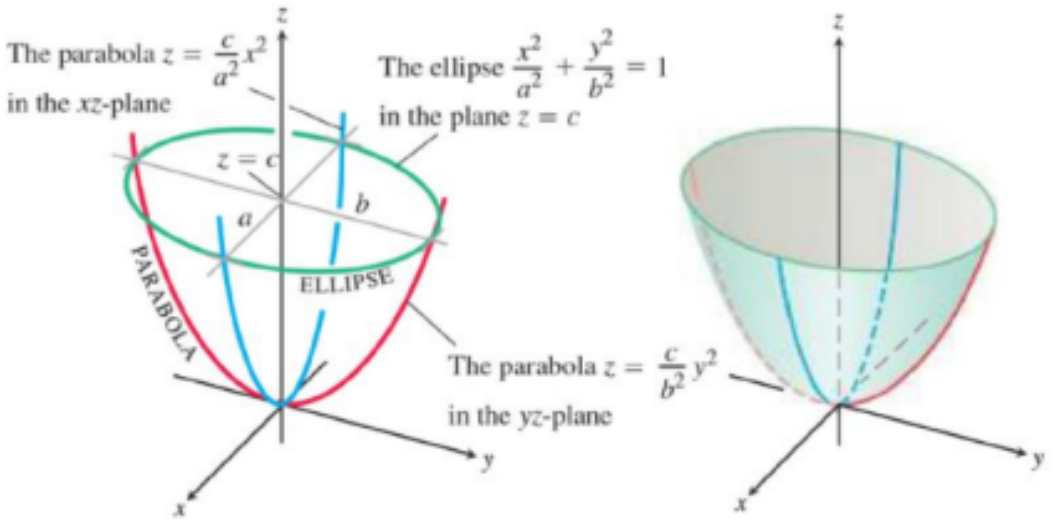
$$Ax^2 + By^2 + Cz^2 + Dxy + Exz + Fyz + Gx + Hy + Iz + J = 0$$

where $A, B, C, D, E, F, G, H, I, J$ are constants.

Ellipsoid



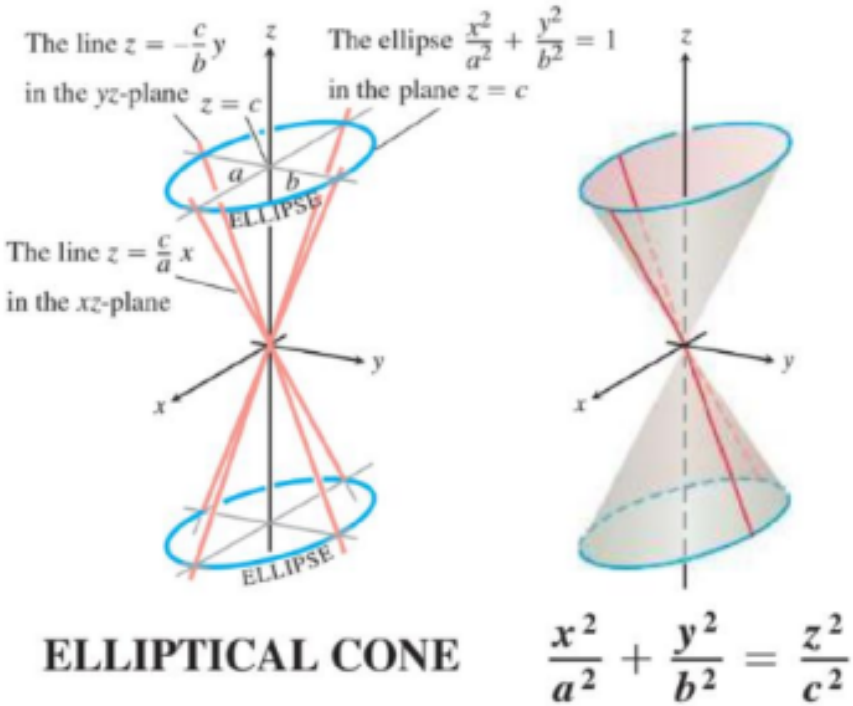
Elliptical Paraboloid



ELLIPTICAL PARABOLOID $\dfrac{x^2}{a^2} + \dfrac{y^2}{b^2} = \dfrac{z}{c}$

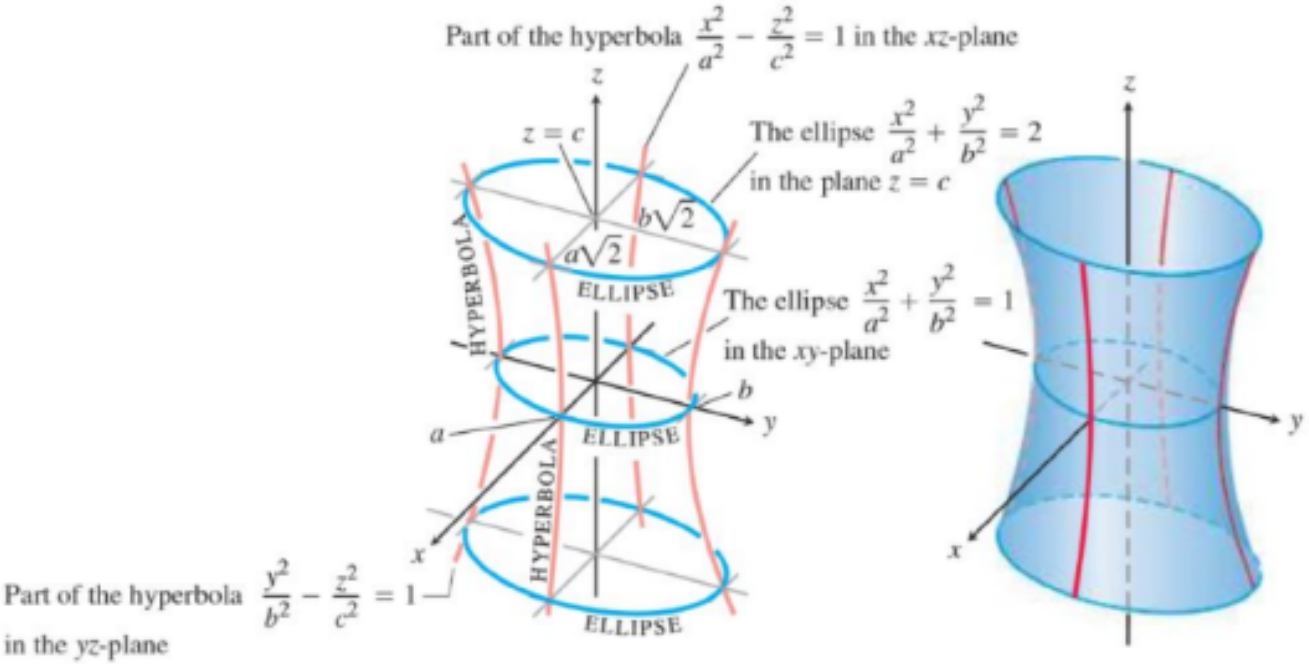
Quadric Surfaces

Elliptical Cone



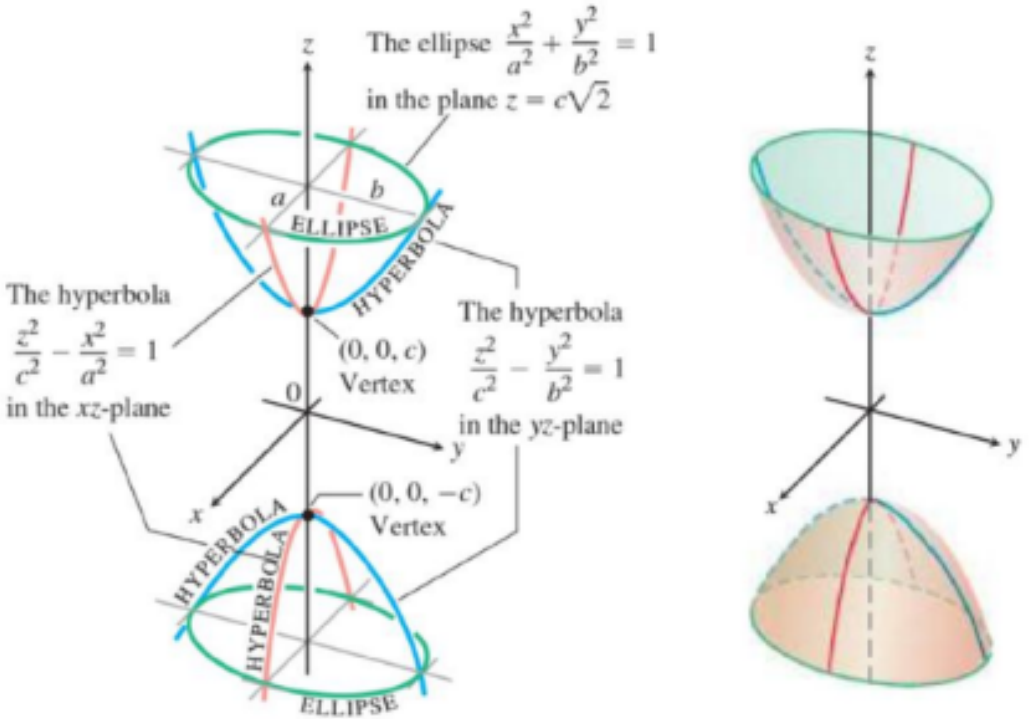
ELLIPTICAL CONE $\quad \dfrac{x^2}{a^2} + \dfrac{y^2}{b^2} = \dfrac{z^2}{c^2}$

Hyperboloid of One Sheet



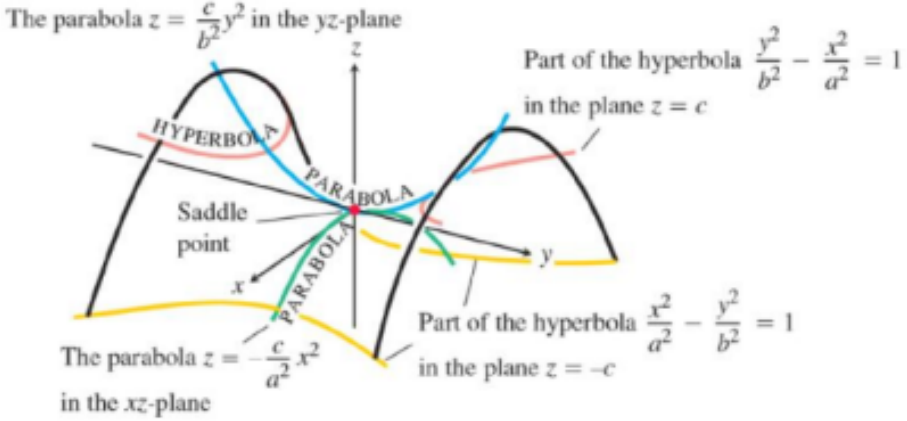
HYPERBOLOID OF ONE SHEET $\quad \dfrac{x^2}{a^2} + \dfrac{y^2}{b^2} - \dfrac{z^2}{c^2} = 1$

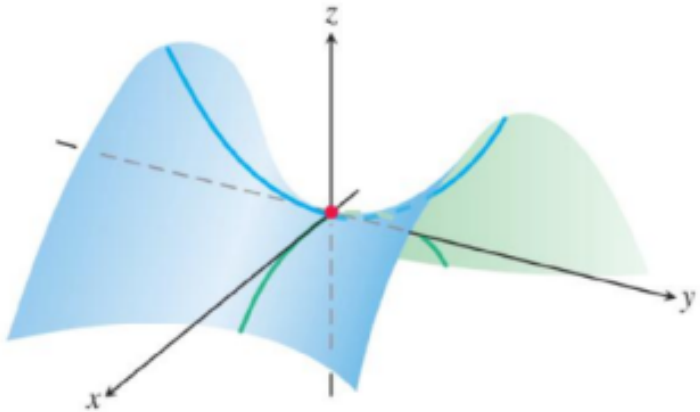
Hyperboloid of Two Sheets



HYPERBOLOID OF TWO SHEETS $\frac{z^2}{c^2} - \frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$

Hyperbolic Paraboloid





HYPERBOLIC PARABOLOID $\frac{y^2}{b^2} - \frac{x^2}{a^2} = \frac{z}{c}, \; c > 0$

Quadric Surfaces

Example: Identify the surface given by the equation

$$x^2 + y^2 + 4z^2 - 2x + 4y + 1 = 0.$$